

Phase 4.3 V2 vs V3 Matrix Analysis Report

1. Overview of Phase 4.3 Experiment

Phase 4.3 evaluated the addition of workload-aware features in experimental V3 models compared to the frozen V2 baseline. All evaluations were conducted using 947 real telemetry records in an isolated shadow mode.

2. Matrix Analysis & Scores (All Workloads)

Model	Precision	Recall	F1	FPR	Latency (ms)
V2_VOLUME_MAD	0.5556	0.1579	0.2459	0.0842	0.0157
V2_VOLUME_EWMA	0.4706	0.0211	0.0403	0.0158	0.0160
V2_VOLUME_CUSUM	0.5714	0.0211	0.0406	0.0105	0.0149
V2_OPERATIONAL_ISO	0.4609	0.1553	0.2323	0.1211	0.0055
V2_OPERATIONAL_LOF	0.7444	0.4368	0.5506	0.1000	0.0057
V2_OPERATIONAL_SVM	0.4933	0.0974	0.1626	0.0667	0.0042
V3_VOLUME_MAD	0.3602	0.3289	0.3439	0.3895	0.0000
V3_VOLUME_EWMA	0.5714	0.0211	0.0406	0.0105	0.0000
V3_VOLUME_CUSUM	0.4837	0.1947	0.2777	0.1386	0.0168
V3_OPERATIONAL_ISO	0.3571	0.0263	0.0490	0.0316	0.0043
V3_OPERATIONAL_LOF	0.9362	0.3474	0.5067	0.0158	0.0030
V3_OPERATIONAL_SVM	0.3975	0.5921	0.4757	0.5982	0.0021

3. Evaluation Conclusion

VOLUME Domain: V2 CUSUM vs V3 Volume Models

V3 Volume models attempted to use normalized z-scores to improve recall. While recall improved, FPR drastically degraded (13.8%), violating the 5% maximum threshold. V2 CUSUM remains the active model due to its robust 1.0% FPR.

OPERATIONAL Domain: V2 LOF vs V3 LOF

V3 LOF introduced normalized throughput features instead of raw duration. This created a massive FPR reduction from 10.0% to 1.5%, while maintaining exceptional precision (93.6%). V3 LOF is a clear candidate for promotion.

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DISTRIBUTION Domain:

INSUFFICIENT_DATA - The real telemetry lacked distributional aggregates.